

from WinFS is Virtual Folders. To understand Virtual Folders, think of a music collection that is stored together on the hard disk physically—there won't be any folders. However, when you open the collection, the files will be grouped according to artist name. You can also choose to view by album name, and selecting an album will list music files with that respective album name. This is basically grouping the files based on certain metadata. For pictures, users can add tags to describe a picture—one picture can have more than one tag. Using this tag, pictures can be grouped and folders created on the fly. This is not a new idea—it is used in music software such as Windows Media Player.

Learning in search

Search engines may learn by observing the user's interaction patterns, and behave accordingly. One approach is to provide the engine with both positive and negative feedback, based upon its performance. For example, if a search on 'orchids' returns forty results, and the user clicks on one of the results, this is noted in an index. Later, if someone does a search on a related term, say 'flowers', the results that have a high score for 'orchids' are returned first. This needs a database of related searches. On

the other hand, if no results in the first search were clicked, all the results get a negative ranking.

Stephen Richardson of Microsoft says that clever searching facilities can be embedded into personal software. He suggests that, for example, while you're writing a document in your word processor, "It could be, automatically or at some prompt from you, searching for material on your disk that relates to what you're writing about. The trick here is that you will not need to formulate a query, because the text you are writing will serve that purpose." He goes on to say that such a system might learn in other ways too, for example, learning about what kinds of source you tend to consult for what purposes.

A new search frontier

WebFountain is a search platform that IBM researchers have been working on for some years, and it may be available soon. It will not be a free search engine—it will be a paid service aimed primarily at businesses and corporations.

WebFountain aims at gathering intelligent and contextual information—going beyond mere keyword search. After crawling the Net, WebFountain will be able to bring back

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Jargon Buster

Semantics: 'Semantics', put simply, means 'meaning'. Thus, 'the semantics of a computer program' would refer to what the program means, rather than what language it is written in, and so on. 'The semantics of search', therefore, refers to search engines and crawlers being aware of the meaning of the data they are dealing with, rather than treating it as mere strings.

Free-format Query: A query expressed in natural language, as opposed to, for example, a boolean query like ('flowers' OR 'orchids') AND NOT ('roses').

Crawlers: The programs that most search engines use to document the Web. Crawlers visit Web sites to read the information in them, and follow the links that the Web site points to. The information that the crawler gathers is sent back to a central database, where it is indexed. This indexing enables the pages to be retrieved upon demand.

Natural Language Processing: The technology that processes strings expressed in natural, or human languages such as English, and turns them into something machine-understandable. For example, 'the dog eats the frog' would be converted into a machine representation that incorporates the concepts of dogs and frogs, the activity of eating, and so on.

Keywords: Keywords and Key-phrases are words and phrases, respectively, that describe, or point to, what a document or Web site is about. For example, the keywords and key-phrases for this article would be 'search', 'search engines', 'information retrieval', and so on.

Term Document Matrix: A TDM is simply a matrix with 0s and 1s, where a 0 means that the keyword defining the row does not appear in the document defining the column, and a 1 means that it does.